

Figure 4: Analysis of the different security risks for Docker containers (Image source: [googleimages.com](#))

the time that any hacker has to exploit a running application in a container can be minimised.

- Some container management systems have also created associated encryption and security services like Docker Secrets to tackle different security concerns associated with containerisation. The practice of using kernel modules, which isolate different processes, also protects containers. This helps to better manage the way distributed applications can access and transmit data.

Advantages and drawbacks of containerisation

Advantages

- Containers used in containerisation are much more lightweight as they use much fewer resources than virtual machines.
- Containerisation helps to increase the efficiency of CPUs by saving the storage and memory of the system.
- Containerisation leads to high portability.
- With the help of containerisation, different containers can be instantiated precisely when they are required, and they can disappear when no longer required. This helps in freeing up the resources on their hosts when they are not in use.
- A single server can be used to host a far greater number of containers than virtual machines and, hence, supports sharing.
- We can track successive versions of a container, inspect the differences, or even roll back to the previous versions using containerisation.
- Containerisation leads to greater modularity.
- It helps us make different changes to the various modules of the application without us having to work on the entire application.

Drawbacks

- Containerisation suffers because it is not adequately isolated from the core OS.
- Since application containers are not abstracted from the host OS on a VM, security threats can access the entire system.
- Containerisation is ideally used for microservice

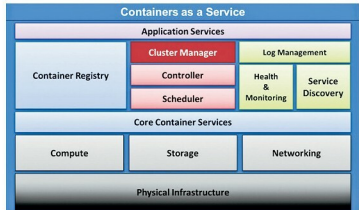


Figure 5: Different types of container services (Image source: [googleimages.com](#))

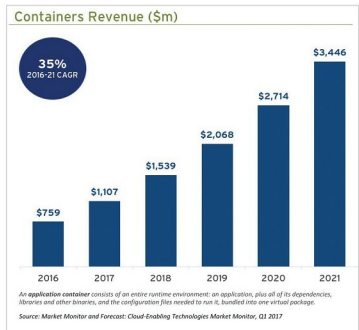


Figure 6: Expected growth in revenue generation due to containerisation (Image source: [googleimages.com](#))

applications. Containers cannot be used as a universal replacement for VMs.

- Few dependencies on the containers limit their portability between servers.
- There are a limited number of tools available that can monitor and manage different containers. 

References

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